

PRANAY TELUKUNTLA

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Professional Experience

Graduate Research Assistant, *The University of Texas at Arlington*

Jun 2025 – Present

- Enhanced network predictive analytics efficiency by 20% by developing and optimizing real-time inference algorithms using Python and PyTorch on live data streams.
- Facilitated collaboration and knowledge transfer by preparing 5+ technical presentations and detailed documentation for academic and industry stakeholders.
- Increased experimental validation success rate by 30% through close collaboration with supervisors on experimental design and integration of LLM outputs into wireless system frameworks.

Data Analyst - Mail Services and Business Affairs, *University Of Texas At Arlington, USA*

Oct 2023 – May 2025

- Developed internal tools using Microsoft Access and VBA to automate tracking and reporting of student service forms, reducing manual processing time by 60% and improving data accuracy.
- Engineered a data pipeline to generate daily financial and operational reports, integrating Excel macros and SQL queries to consolidate postage sales and department-level analytics for decision-making.
- Built an asset inventory dashboard to manage and log the onboarding of over 300+ electronic devices across university departments, enhancing traceability and compliance reporting.
- Scraped pricing data from postage and packaging vendors using Python (BeautifulSoup, Selenium) to compare rates and negotiate better contracts, achieving a 10% reduction in procurement costs.

Software Engineer – Developer, *High Radius Corporation, India*

Jun 2022 – Apr 2023

- Designed and deployed distributed validation pipelines across multi-region dev/test/prod clusters, reducing deployment errors by 25% and ensuring ML model retraining pipelines operated with 99% reliable.
- Developed scalable PL/SQL ETL workflows processing over 50k+ records weekly, ensuring data consistency for ML analytics.
- Improved nightly batch job completion times by 35% via query optimization and resource-parallelization, enabling faster time-to-decision in financial reconciliations.
- Led end-to-end QA and post-deployment model monitoring across 15+ releases, incorporating model evaluation, feature drift detection, and alerting.

Skills

Languages: Python, SQL, JavaScript, Bash, MATLAB

ML Frameworks: scikit-learn, XGBoost, PyTorch, TensorFlow, LangChain, OpenCV

Data/Tooling: NumPy, pandas, FAISS, Jupyter, Google Colab, VS Code, REST APIs, ETL Pipelines

Cloud & DevOps: AWS (S3, EMR, Lambda), OCI, Git, Linux, Docker, CI/CD, Airflow (learning)

ML Concepts: Supervised Learning, Recommender Systems, Distributed ML, Parallel Training, Model Monitoring, Feature Drift, RAG, SHAP, Vector Search, Optimization, Feature Engineering, Model Explainability

Certifications: AWS Certified Data Engineer, Agentforce Specialist

Education

University of Texas at Arlington, *Master of Science in Computer Science*

Aug 2023 – May 2025

- Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Mining, Numerical Methods, Software Engineering
- Specializations:** **Big Data Management & Intelligent Systems** GPA: 3.55/4.0

Research & Projects

LLM-Guided Navigation in Millimeter Wave Network (Ongoing)

PyTorch, GPT-4o, Dijkstra Algorithm

- Benchmarked **fine-tuned GPT-4o** against **base GPT-4o** and traditional Dijkstra algorithm for optimal signal strength pathfinding in simulated mmWave environments.
- Demonstrated LLM-guided routing reduced path computation time by **21%** while maintaining signal integrity across **large-scale dynamic network topologies**.

Amazon Product Recommender

scikit-learn, TF-IDF, pandas

- Achieved **90%** precision@10 using hybrid **TF-IDF + collaborative filtering recommenders**, improving retrieval for personalized product feeds.
- Engineered **scalable similarity pipelines** for **10K+ products**, driving **22% engagement uplift** via optimized ranking scores and feature tuning and implemented recommender model retraining pipelines with drift-aware feature updates.

Telecom Churn Predictor

scikit-learn, XGBoost, Python

- Improved churn classification accuracy by 15% using stratified sampling and cost-sensitive **XGBoost**, enabling actionable insights for customer retention. Deployed trained model into simulated production pipeline with model versioning and monitoring.
- Delivered feature explainability via **SHAP**, supporting business-driven retention programs with **11% projected ROI lift**.

Collaborative Filtering Recommender System

Python, NumPy, pandas

- Boosted ranking precision by **18%** using **matrix factorization with regularization**, optimizing recommendations in cold-start conditions and constructed **cosine-based affinity matrices** across **thousands of users**, reducing fallback errors by **20%**.

E-commerce & Reactive Microservices Platform

Spring Boot, WebFlux, Kafka, Docker, Liquibase, CI/CD

- Developed an end-to-end platform combining e-commerce microservices and reactive demos, achieving a **35%** improvement in scalability. Enhanced inter-service communication via API Gateway and Eureka, reducing response latency by **30%** and boosting throughput by **25%**. Streamlined deployments with Docker, Liquibase, and CI/CD pipelines, cutting update cycles by **40%** and significantly increasing system responsiveness and reliability.